



California State University, Long Beach

AIAA 2024-2025 Design, Build, Fly Proposal



1. Executive Summary

This proposal outlines California State University, Long Beach DBF team's plan to design, build, and fly an aircraft for the AIAA 2024-2025 Design, Build, Fly competition. This year, the aircraft will deploy a small autonomous aircraft mounted on the underside of the main vehicle and carry two bottles, no smaller than 16 fl oz each, as fuel tanks with as much weight as possible. The main aircraft must have a maximum wingspan of 6 feet and its propulsion battery energy cannot exceed 100 Watt-hours. The X-1 test vehicle, the aircraft mounted on the underside of the main vehicle, must not exceed 0.55lbs and must land in a 200 by 200 foot box adjacent to the runway. The total score, and therefore our design requirements, is determined by our payload capacity, aircraft speed, propulsion system energy capacity, and the accuracy of the X-1 test vehicle.

Through sensitivity analysis, we determined that maximizing payload weight and minimizing lap time on the main aircraft was the optimal method to increase score, where stall velocity is the main constraint of payload weight. To maximize payload weight, the full 6 feet of wingspan will be utilized, with about 8 square feet of total wing area. A flying wing configuration will be used for both the main aircraft and the glider, as it has less drag, and payloads will be mounted on pylons under the wing regardless. The aircraft will use a single two blade puller propeller, a NACA 24112 airfoil, and retractable tricycle-configuration landing gear. The fuel tanks will be mounted near the ends of the wings to optimize wing bending stress, and the X-1 test vehicle will be mounted directly under the center of gravity to avoid changing the flight characteristics of the main aircraft after deployment. The project timeline has been created to include time for autopilot, propulsion, structural, and flight testing, with prototype construction beginning in mid november. The prototype will be flown and optimized for score, through continuous design iterations, before the competition aircraft design is finalized.

2. Management Summary

2.1. Administration & Organization

Long Beach Aviation (LBA) represents the CSULB AIAA student branch and is the only aircraft project club on campus. The purpose of this organization is to give student members the opportunity to develop their technical skills and apply them to aircraft design competitions. This year, the club consists of 30 student

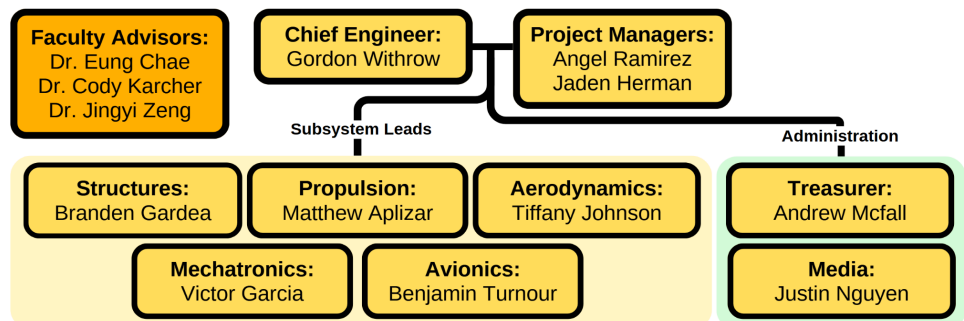


Figure 2.1: Organization Flow Chart

members, 3 faculty advisors, 5 subsystems, and three club leaders shown in figure 2.1. The Project Managers provide development resources to each subsystem, organize meetings with the team, handle the administrative and organizational functions of the club, and monitor the overall progress of the project. The Chief Engineer is responsible for leading the subsystem teams in the technical aspects of aircraft development and ensuring the integration of subsystems meet quality requirements. Together the project manager and chief engineer relay overall progress to advisors and members as the joint club leaders.

Subsystem leads are responsible for their respective aircraft subsystems, which includes coordination of work, monitoring team member progress, and scheduling tasks and meetings. Subsequently all subsystems collaborate with each other and the Chief Engineer to integrate design features and sizing calculations. The Pilot is responsible for the operation of the aircraft during test flights and competition flight missions, and also actively participates in design and development. A descriptive explanation of the roles and required skills of every subsystem lead is shown in Table 2.2.

Team	Roles	Required Skills
Aerodynamics	Select optimal airfoils and analyze performance of design iterations using CFD software.	Experience with CFD, MATLAB, Excel, aircraft optimization and flight dynamics.
Avionics	Research and configuration of avionics suites, to stabilize carrier A/C and control glider detach	Aircraft stability, embedded systems, electronics fundamentals
Structures	Design, manufacturing, and testing of airframe components and payload systems.	Experience with Solidworks CAD, MATLAB, and computational structural analysis, or FEA
Propulsion	Optimal powertrain sizing and selection using manufacturer data and thrust stand testing.	Experience in MATLAB or Python and brushless motor electronic
Mechatronics	Develop electronically controlled mechanical systems and improve manufacturing processes.	Embedded systems and electronics fundamentals. Proficiency in CAD and design of mechanical systems

Table 2.2: Subsystem Requirements and Skills

2.2. Project Timeline

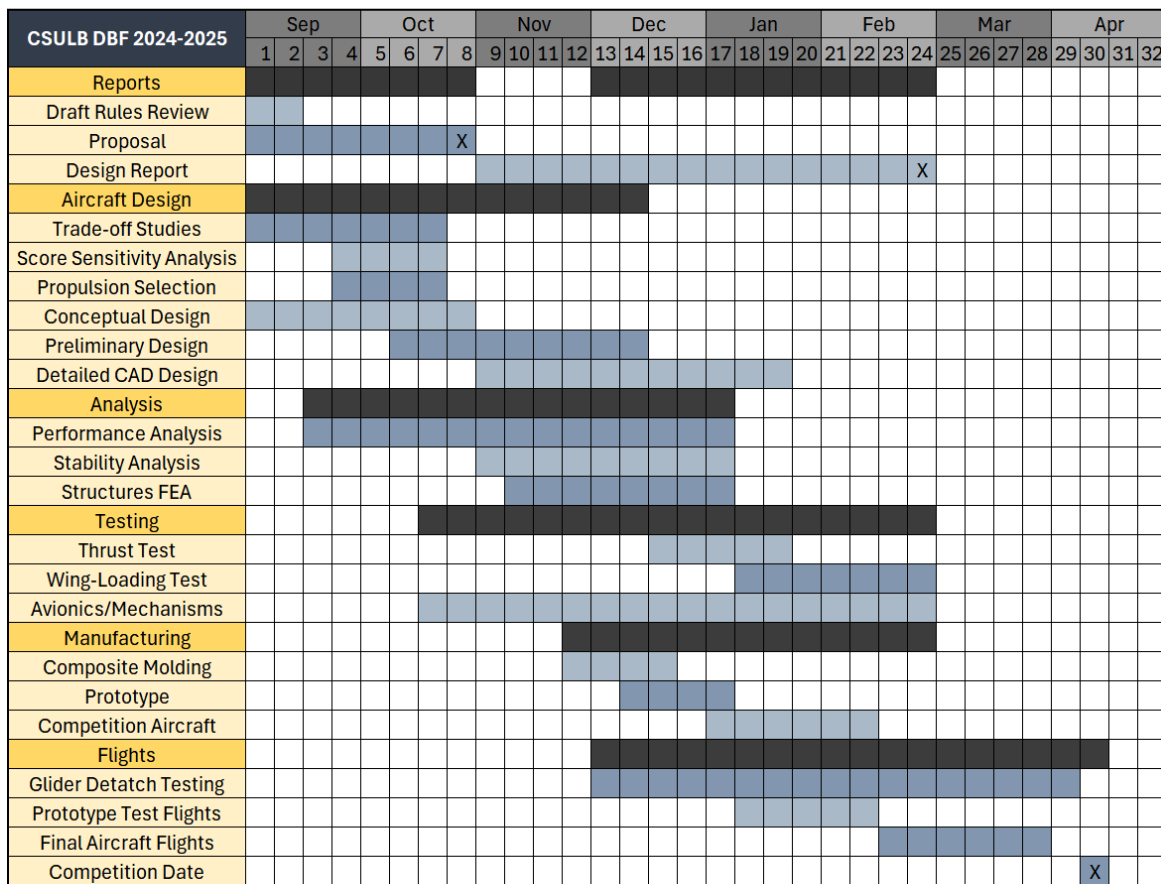


Figure 2.3: Gantt Chart

Figure 2.3 illustrates a detailed project timeline encompassing key milestones across all phases of the design process, beginning with the release of the AIAA DBF 2024-2025 draft rules and culminating to the competition date. The club leadership holds weekly meetings with members and advisors to establish goals and assess progress, ensuring that the leadership team remains aligned on priorities and maintains a unified vision. Attendance at the weekly General Body Meetings (GBM), where the majority of work takes place, is expected of all members. In addition, every technical lead conducts their own weekly meetings to engage in more detailed discussions and focus on specific tasks.

2.3. Budget

The sum of the anticipated costs includes aircraft structural materials, electronics, and travel expenses as shown in Table 2.4. The aircraft structural costs come from the intention to use sandwich composite, resulting in a large cost. The electrical category is split between our propulsion, avionics, and mechatronics subsystems. The travel category includes fuel because we plan on driving to the competition. The total project costs, including travel and materials, will be covered by academic and industry grants, in addition to funding from the College of Engineering at California State University, Long Beach.

Category	Item	Cost
Structures (\$2,315)	Carbon fiber, fiberglass	\$900
	Core material	\$850
	Resin pump set	\$225
	Mechanical hardware	\$340
Electronics (\$2,180)	Software	\$260
	Servos, receiver, controller	\$520
	Batteries, battery charger	\$1,050
	Motors, propellers, LED's	\$350
Travel (\$3,380)	Housing	\$1,300
	Fuel	\$880
	Food	\$1,200
Anticipated Total Cost		\$7,875

Table 2.4: Budget Summary

3. Conceptual Design Approach

3.1. Mission Requirements

Mission	Scoring	Objectives	Design Requirements
M1	= 1	- Complete 3 laps - Land successfully	The carrier aircraft must take off, fly, and land safely under its own power.
M2	= 1 + $\left[\frac{(\frac{\text{Payload Weight}}{\text{Lap Time}})_n}{(\frac{\text{Payload Weight}}{\text{Lap Time}})_{\max}} \right]$	- Minimize lap time - Complete 3 laps - Maximize payload weight	The carrier aircraft must withstand loads associated with increased airspeed, lift, and thrust.
M3	= 2 + $\left[\frac{(\#Laps + \frac{\text{Bonus Score}}{\text{Glider Weight}})_n}{(\#Laps + \frac{\text{Bonus Score}}{\text{Glider Weight}})_{\max}} \right]$	- Maximize laps flown in five minutes - Successfully land glider in bonus box - Minimize glider weight - Complete a successful landing	Carrier aircraft must fly for the full length of the mission. Glider must reliably transition to stable straight and turning flight after detach. Minimize glider descent time.
Ground Mission	= $\frac{(\text{Mission Time})_{\min}}{(\text{Mission Time})_n}$	- Rapid pylon and fuel tank installation - Successfully launch the glider	Minimize loading time for both our payload and glider through quick attachment mechanisms.

Table 3.1: Mission Scoring Equation Summary

The objective of this year's DBF competition consists of three flight missions and a single ground mission. The mission scoring equations are presented in table . Along with the scoring equations, the objectives and general requirements are documented in the table below. Upon the review of all flight mission scores, common objectives are observed. Payload weight and flight speed are the two important flight characteristics that are prioritized during scoring analysis.

The ground mission specific design decision was to minimize the fuel tanks to the competition constraint of two to ensure a quick and efficient loading time is obtained. In order to maximize M2 and M3 flight time for a given power setting, the battery energy was maximized at 99.9 Wh. The goal of mission 2 is to maximize the propulsion system based on dash performance. Our estimated flight time for 3 laps, including a safety factor, determined our propulsion system average power draw to be 2600 W. This power limit was then used to recalculate our lap time giving an approximate lap time of 26.1 seconds for mission 2. The goal for mission 3 is to ensure a total flight time as close to 5 minutes as possible. The mission approach is to decrease glider descent time with an initial estimate of 30 seconds. These estimates led to a mission 3 power draw approximate of 1300 kW for a 270s flight time.

3.2. Sensitivity Analysis

Aircraft drag was calculated from total weight and wing area estimates. Total weight was found to be a function of both wing area and payload weight. The total weight function was derived from empirical structural sizing data from previous team aircraft. Aircraft drag and thrust available were plotted with respect to velocity. Thrust curves were created with APC prop data, for a feasible motor and propeller combination given specific power settings. These power settings were chosen to sweep through the M3 and M2 fixed average power draw setting. The thrust required curves were traded with payload weight for the specific missions; M2 has 12 lbs of payload while M3 has no payload. The intersections of the thrust available and thrust required (drag) curves specify maximum dash velocities for specific thrust available settings. The team observed that aircraft dash velocity was largely dependent on power setting, and insensitive to change in payload weight, especially at higher airspeeds. Different propeller thrust curves were analyzed in order to maximize airspeed at the M2 power setting, and provide acceptable thrust for the M3 reduced power setting. M3 feasibility is determined through comparing the required thrust with no payload to the M3 power setting thrust curve. For an M3 power draw of 1300 W and a corresponding airspeed of about 128 ft/s, the M3 average lap time was estimated with flight dynamics equations to be 33 seconds, giving a total number of laps of 8.

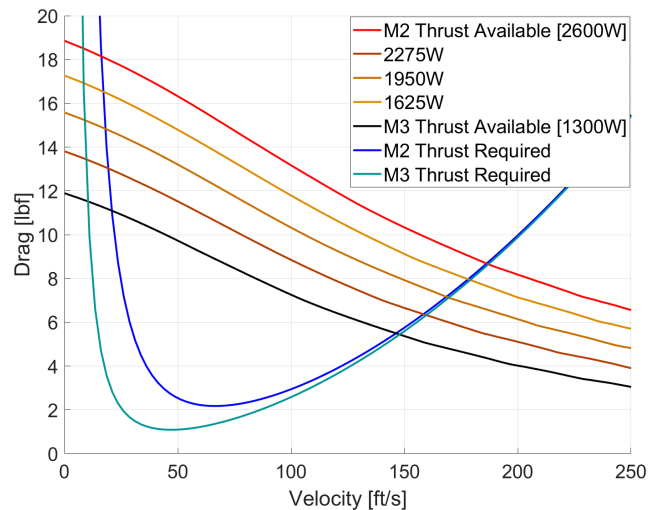


Figure 3.2: Drag vs Velocity For Various Power Settings

M2 change in relative score was plotted as a trade sweep of payload weight and aspect ratio. The drag values, calculated for each payload weight and aspect ratio, and the M2 thrust required curve, gained from figure 3.2, were matched to solve for dash velocity. The M2 score was plotted as contours, and was found to generally increase with higher aspect ratios and payload weights. As the team aimed to prioritize the pilot handleability of the aircraft, a level flight stall speed limit was developed to restrict the aircraft's wing loading to feasible values. Stall speeds of 40-50 ft/s were swept through to determine sensitivity on score. The team determined the upper limit to be 50 ft/s for safe takeoffs and landings. Curves of high stall speeds showed optimum ranges of AR and Wpl, where the relative score would decrease in either direction from the optimum region. This optimal region was seen to shift towards higher scores at higher payload weights and lower aspect ratios for higher stall speeds. For the 50 ft/s stall speed curve, the optimum combination of an AR = 4.5 and Wpl ~ 12lb was shown to produce the highest possible M2 relative score. Due to this combination of aspect ratio and payload weight only being optimal for the thrust curve of a specific propeller, the propulsion system is optimized through plugging these variables back into figure 3.2 to calculate required thrust values, where different propellers are analyzed for their M2 dash speed and M3 feasibility.

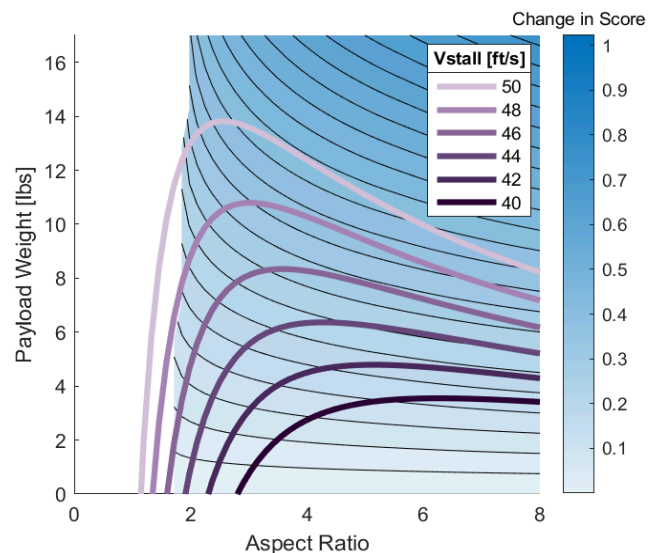


Figure 3.3: Mission 2 Scoring Parameter Analysis

3.3. Aircraft Preliminary Design

A flying wing configuration was selected in order to lower the parasitic drag coefficient, maximize dash airspeed, and increase the structural efficiency of the airframe for higher payload capacity. Due to internal payload not being required, a conventional fuselage was omitted in favor of an extended root chord to store only necessary flight electronics inside the wing. Extending the root chord increases total thickness, allowing for a lighter overall wing structure with respect to bending stress. The primary objective of static longitudinal stability determines root-to-tip sweep and twist angle. The secondary objective of aircraft lateral-directional stability is achieved through sizing vertical tail surfaces, which will be optimized through the use of inviscid vortex-lattice-method calculations. The initial design solution has been found to have a wingspan of 72 in, an aspect ratio of 4.5, a root-to-tip sweep angle of 32° , a wingtip twist of 6° , and a total vertical tail area of 0.29 sqft. This aircraft's expected minimum weight is 11.8 lb and 23.8 lb loaded. A tractor propeller configuration was selected to balance the CG forward, utilize high propulsive efficiency, and to allow for propeller clearance during takeoff rotation. The initial airfoil was selected as a NACA 24112 due to its moderately high lift coefficient while maintaining a negative moment coefficient. This allows for an increased lift to drag ratio due to lower required geometric twist to maintain pitch stability. Two rudders on the vertical tails ensure yaw control during takeoff and landing. The carrier aircraft will use elevon control surfaces positioned near the wingtips to control pitch and roll axes.

Due to the battery power constraint, a 12S 99.9 Wh battery was selected. The battery is joined with a 16x16 propeller and a motor with a kv of 4500 from T-motor to complete the powertrain of the carrier aircraft. Pylons will be made from semi-rigid 3DP plastic, and will include an over-center mechanism to compression-hold the fuel tanks in flight. Pylons will be attached to the airframe using quick-connect fasteners to decrease time spent loading the carrier aircraft. The glider will be mounted on a single pylon and will use a static connector that interfaces with a servo-actuated detach mechanism within the main carrier aircraft. The glider configuration was selected to be a low aspect ratio flying wing to reduce the structural weight required to prevent flutter while in the propeller slipstream, which would limit the maximum airspeed of the carrier aircraft.

4. Manufacturing Plan

Illustrated in Figure 4.1 is the preliminary manufacturing flowchart, encompassing the entire process from conceptual and preliminary design to final aircraft manufacturing. It details the main manufacturing methods, including laser cutting for various structural members, CNC milling of foam molds for composite sandwich structures and wet processes, and 3D printing for pylons, fixtures, and mounts.

4.1. Critical Manufacturing Process

For the initial materials selection, the internal structure of the wing will use spars and ribs manufactured from composite foam-core panels cut on a custom CNC Hotwire, and the wing skins will make use of a carbon fiber and fiberglass reinforced sandwich structure, with a core material conforms to the part curvature. The aerodynamic surfaces will be manufactured similarly to the wing skins. For the composite parts, the matrix will use marine-grade epoxy due to its accessibility and excellent strength properties. For aircraft hard-points, high strength laser-cut aircraft plywood will be utilized. All internal structure parts will be bonded to each other and the skins using a mixture of epoxy and filling agent.

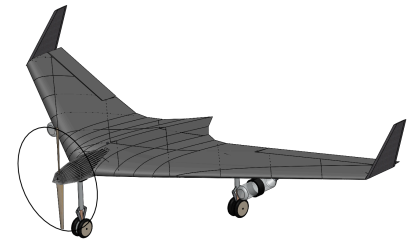


Figure 3.5: Aircraft Model Trimetric View

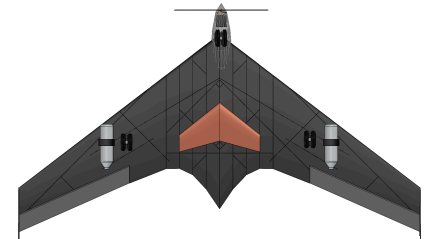


Figure 3.4: Aircraft Model Bottom View

In order to facilitate repeatable manufacturability for both the prototype and competition aircraft, the manufacturing process for the aircraft skins will use composite tooling manufactured from CNC-cut foam. All composite components will be manufactured through wet-layup and vacuum bagging processes.

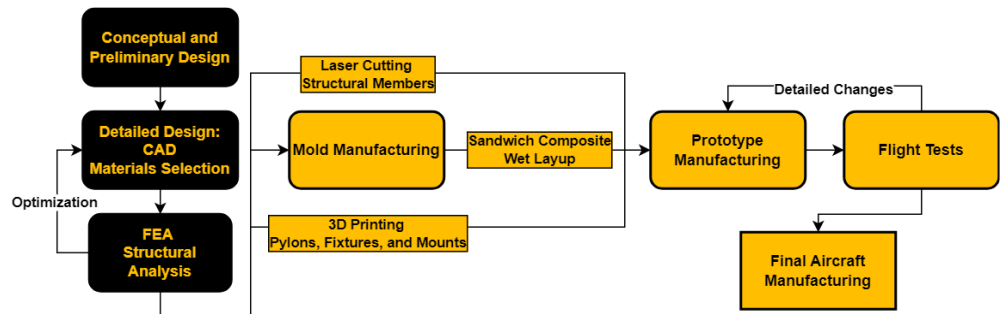


Figure 4.1: Manufacturing Flow Chart

5. Testing Plan

Multiple tests will be run before the construction of the final aircraft, to ensure that our calculations and assumptions hold true. The motor will be run on a test stand both statically and dynamically, to ensure that it produces the calculated amount of thrust. The wing spar will be tested by mounting weights on the tips and measuring the deflection. Once the aircraft is built, landing gear tests will be performed by putting the aircraft in its fully loaded configuration and dropping it from a de red sired height. After testing radio connection and failsafe reliability, the vehicle will be cleared for flight testing. Taxi and take off tests will be performed to confirm the calculated values, or identify areas of improvement. Multiple full mission flight tests will then be performed to fine tune the flight performance of the aircraft, and optimize for score. The glider development will be run separately, with autopilot tests done on a separate test aircraft to remove dependency on other subsystems. Before the full mission test flights, glider release tests will be performed to ensure that the release is smooth and consistently activates the autopilot protocol.

Test	Objective	Method	Sub-System
Preliminary Tests			
Static Thrust Test	Test different motor and prop combinations to verify it produces the expected thrust.	Use a thrust test stand to collect thrust, torque, battery current, and RPM of different motor and propeller combinations.	Propulsion
Dynamic Thrust Test	Collect thrust test stand data in freestream air to get empirical thrust vs. velocity curves.	Mount the test stand in a wind tunnel and collect data on different motor and prop configurations at different airspeeds.	Propulsion
Wing Load Test	Verify maximum load factor of aircraft wing structure to simulate a distributed wing load.	Mount center of aircraft on fixture, load underside of wing with spanwise weights matching lift distribution at max g-loading.	Structures
Wingtip Test	Verify that the aircraft passes the 2.5g competition wing loading requirement.	Load the aircraft with the maximum payload weight and mount it on the wingtips, and measure the deflection.	Structures
Ground Testing			
Landing Gear Stress Test	Verify that the landing gear withstands takeoff and landing shock loads.	Drop the aircraft from an appropriate height. The propeller must not strike the ground at full suspension bottom-out.	Structures/ Mechatronics
Failsafe testing	Test to ensure that the correct commands are given by the receiver during a failsafe.	Turn off the radio transmitter while the aircraft is powered, and check to see that the correct failsafe commands are given.	Avionics
Radio Testing	Ensure that the radio is fully functional, providing adequate data to the receiver.	Verify radio transmitter and receiver communicate at a sufficient range, and at all orientations.	Avionics
In Flight Testing			
Take-Off Speed Test	Record and analyze take-off performance data to optimize performance.	Increase throttle to full, rotate at the calculated rotation speed, and compare the liftoff location to the theoretical values.	Aerodynamics
High Speed Taxi Test	Test the nosewheel and yaw control on the ground, as well as landing gear strength.	Use slow-motion cameras to empirically test that the landing gear holds during future takeoffs.	Mechatronics
Mission Flight Tests	Practice all three flight tests and calculate the scores to identify areas for adjustment.	Perform all three missions and determine if changes to throttle settings, payload weights, etc., can improve scores.	Aerodynamics
Glider Testing			
Glider Detach Test	Test the reliability of the detachment method of the glider from the main aircraft.	Test the detach mechanism to ensure the glider detaches smoothly and the LEDs turn on. Repeat the tests in flight.	Mechatronics
Waypoint Mission Test	Perform gliding waypoint missions to gain familiarity and limit test INAV.	Using a separate test aircraft, perform a waypoint mission and landing similar to the path of the glider in mission 3.	Avionics

Table 5.1: Test Plan Summary

6. Executive Summary

This proposal outlines the California State University, Long Beach (CSULB) Design, Build, Fly (DBF) Long Beach Aviation (LBA) team's plan to design, build, and fly an aircraft for the AIAA 2024-2025 Design, Build, Fly (DBF) competition. This year, the aircraft will deploy a small autonomous aircraft mounted on the underside of the main vehicle

and carry two bottles, no smaller than 16 fl oz each, as fuel tanks with as much weight as possible. Based on the AIAA DBF guidelines, Tthe main aircraft must have a maximum wingspan of 6 feet and its propulsion battery energy cannot exceed 100 Watt-hours. The X-1 test vehicle, the aircraft mounted on the underside of the main vehicle, must not exceed 0.55 lbs and must land in a 200 by 200 foot box adjacent to the runway. The total score, and therefore our design requirements, is determined by our payload capacity, aircraft speed, propulsion system energy capacity, and the accuracy of the X-1 test vehicle.

Through sensitivity analysis, we determined that maximizing payload weight and minimizing lap time on the main aircraft was the optimal method to increase score, where stall velocity is the main constraint of payload weight. To maximize payload weight, the full 6 feet of wingspan will be utilized, with about 8 square feet of total wing area. A flying wing configuration will be used for both the main aircraft and the glider, as it has less drag, and payloads will be mounted on pylons under the wing regardless. The aircraft will use a single two- blade puller propeller, a NACA 24112 airfoil, and retractable tricycle-configuration landing gear. The fuel tanks will be mounted near the ends of the wings to optimize wing bending stress, and the X-1 test vehicle will be mounted directly under the center of gravity to avoid changing the flight characteristics of the main aircraft after deployment. The project timeline has been created to include time for autopilot, propulsion, structural, and flight testing, with prototype construction beginning in mid-N november. The prototype will be flown and optimized for score, through continuous design iterations, before the competition aircraft design is finalized.

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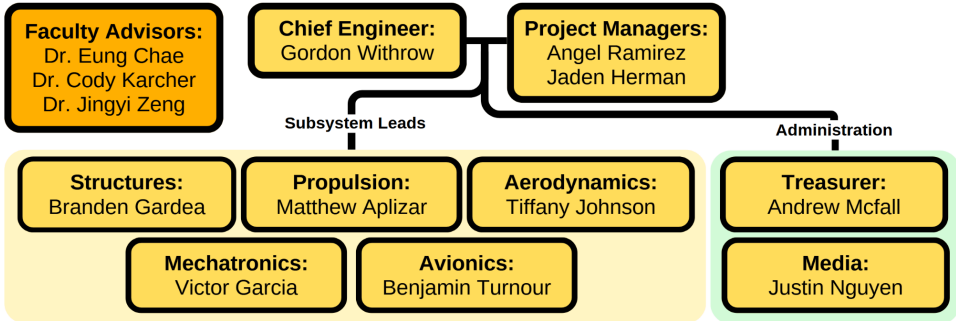


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Figure 2.3 illustrates a detailed project timeline encompassing key milestones across all phases of the design process, beginning with the release of the AIAA DBF 2024-2025 draft rules and culminating to the competition date. The LBAclub leadership holds weekly meetings with members and advisors to establish goals and assess progress, ensuring that the leadership team remains aligned on priorities and maintains a unified vision. Attendance at the weekly General

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The budget summary in Table 2.4 shows the sum of the anticipated costs includes aircraft structural materials, electronics, and travel expenses, as shown in Table 2.4. The aircraft structural costs come from the intention to use sandwich composite, resulting in a large cost. The electrical category is split between our propulsion, avionics, and mechatronics subsystems. The travel category includes fuel because we plan on driving to the competition. The total project costs, including travel and materials, will be covered by associated students inc (ASI) of CSULB, academic and industry grants, in addition to funding from the Mechanical and Aerospace Engineering Department and the College of Engineering at CSULB California State University, Long Beach.

8. Conceptual Design Approach

8.1. Mission Requirements

Mission	Scoring	Objectives	Design Requirements
M1	= 1	- Complete 3 laps - Land successfully	The carrier aircraft must take off, fly, and land safely under its own power.
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M3	$= 2 + \left[\frac{\left(\#Laps + \frac{\text{Bonus Score}}{\text{Glider Weight}} \right)_n}{\left(\#Laps + \frac{\text{Bonus Score}}{\text{Glider Weight}} \right)_{\max}} \right]$	- Maximize laps flown in five minutes - Successfully land glider in bonus box - Minimize glider weight - Complete a successful landing	Carrier aircraft must transition to a stable straight and turning flight after detachment. Minimize glider descent time.
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M2 change in relative score was plotted as a trade sweep of payload weight and wing area. The drag values, calculated for each payload weight and aspect ratio, and the M2 thrust required curve, gained from Figure 3.2, were matched to solve for dash velocity. The M2 score was plotted as contours, and was found to generally increase with lower wing areas and higher payload weights. As the team aimed to prioritize the pilot handleability of the aircraft, a level flight stall speed limit was developed to restrict the aircraft's wing loading to feasible values. Stall speeds of 40-50 ft/s were swept through to determine the sensitivity of the score. The team designated the upper limit to be 50 ft/s for safe takeoffs and landings. The highest possible relative score occurred at high stall speeds. Curves of high stall speeds showed optimum ranges of wing area and payload weight, where the relative score would decrease in either direction from the optimum region. This optimal region was seen to shift towards higher scores at higher payload weights and higher wing areas for higher stall speeds. For the 50 ft/s stall speed curve, the optimum combination of an $AR = 4.5$ and $W_{pl} \sim 12$ lb was shown to produce the highest possible M2 relative score. Due to this combination of aspect ratio and payload weight only being optimal for the thrust curve of a specific propeller, the propulsion system is optimized by plugging these variables back into Figure 3.2 to calculate required thrust values, where different propellers are analyzed for their M2 dash speed and M3 feasibility.

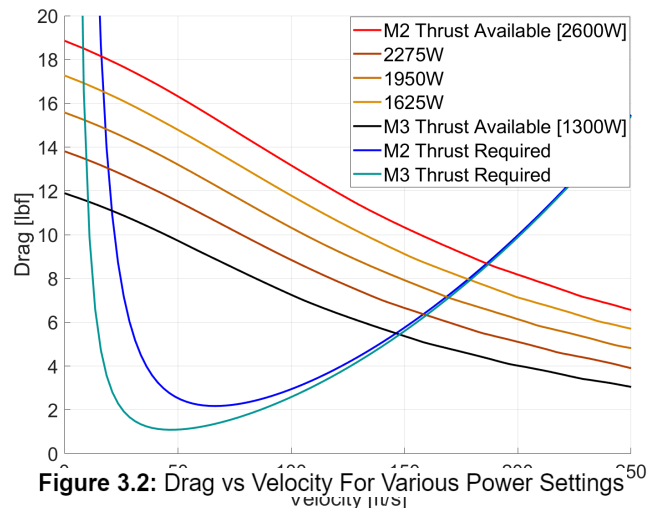
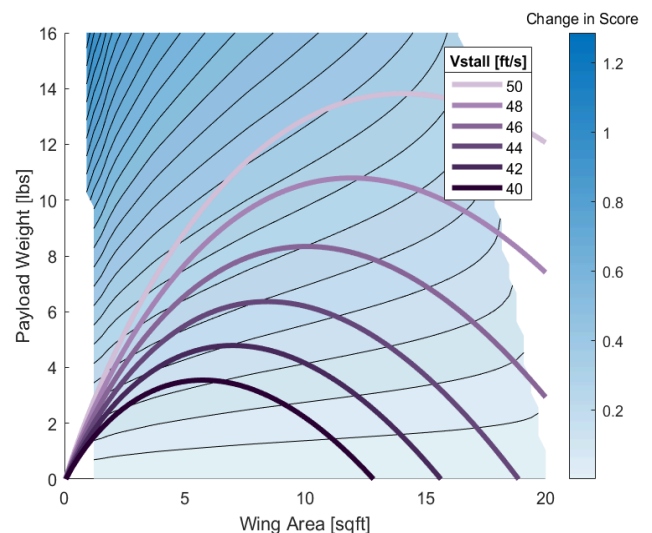


Figure 3.2: Drag vs Velocity For Various Power Settings



8.3. Aircraft Preliminary Design

A flying wing configuration was selected in order to lower the parasitic drag coefficient, maximize dash airspeed, and increase the structural efficiency of the airframe for higher payload capacity. Due to internal payload not being required, a conventional fuselage was omitted in favor of an extended root chord to store only necessary flight electronics inside the wing. Extending the root chord increases total thickness, allowing for a lighter overall wing structure with respect to bending stress. The primary objective of static longitudinal stability is to determine root-to-tip sweep. The primary objective of static longitudinal stability determines root-to-tip sweep and twist angle. The secondary objective of aircraft lateral-directional stability is achieved through sizing vertical tail surfaces, which will be optimized through the use of inviscid vortex-lattice-method calculations. The initial design solution has been found to have a wingspan of 72 in, an aspect ratio of 4.5, a root-to-tip sweep angle of 32° , a wingtip twist of 6° , and a total vertical tail area of 0.29 sqft. This aircraft's expected minimum weight is 11.8 lb and 23.8 lb loaded. A tractor propeller configuration was selected to balance the CG forward, utilize high propulsive efficiency, and to allow for propeller clearance during takeoff rotation. The initial airfoil was selected as a NACA 24112 due to its moderately high lift coefficient while maintaining a negative moment coefficient. This allows for an increased lift-to-drag ratio due to a lower required geometric twist to maintain pitch stability. Two rudders on the vertical tails ensure yaw control during takeoff and landing. The carrier aircraft will use elevon control surfaces positioned near the wingtips to control pitch and roll axes.

Due to the battery power constraint, a 12S 99.9 Wh battery was selected. The battery is joined with a 16x16 propeller and a motor with a kV of 4500 from T-motor to complete the powertrain of the carrier aircraft. Pylons will be made from semi-rigid 3DP plastic, and will include an over-center mechanism to compression-hold the fuel tanks in flight. Pylons will be attached to the airframe using quick-connect fasteners to decrease the time spent loading the carrier aircraft. The glider will be mounted on a single pylon and will use a static connector that interfaces with a servo-actuated detach mechanism within the main carrier aircraft. The glider configuration was selected to be a low aspect ratio flying wing to reduce the structural weight required to prevent flutter while in the propeller slipstream, which would limit the maximum airspeed of the carrier aircraft.

9. Manufacturing Plan

Illustrated in Figure 4.1 is the preliminary manufacturing flowchart, encompassing the entire process from conceptual and preliminary design to final aircraft manufacturing. It details the main manufacturing methods, including laser cutting for various structural members, CNC milling of foam molds for composite sandwich structures and wet processes, and 3D printing for pylons, fixtures, and mounts.

4.1. Critical Manufacturing Process

For the initial materials selection, the internal structure of the wing will use spars and ribs manufactured from composite foam-core panels cut on a custom CNC Hotwire, and the wing skins will make use of a carbon fiber and fiberglass reinforced sandwich structure, with a core material that with a core material conforms to the part curvature. The aerodynamic surfaces will be manufactured similarly to the wing skins. For the composite parts, the matrix will use marine-grade epoxy due to its accessibility and excellent strength properties. For aircraft hard-points, high-strength

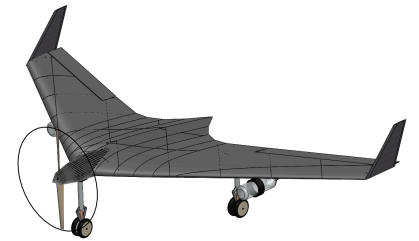


Figure 3.5: Aircraft Model Trimetric View

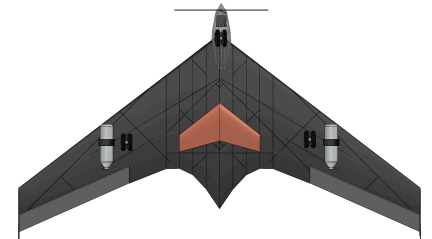


Figure 3.4: Aircraft Model Bottom View

laser-cut aircraft plywood will be utilized. All internal structure parts will be bonded to each other and the skins using a mixture of epoxy and filling agents. In order to facilitate repeatable manufacturability for both the prototype and competition aircraft, the manufacturing process

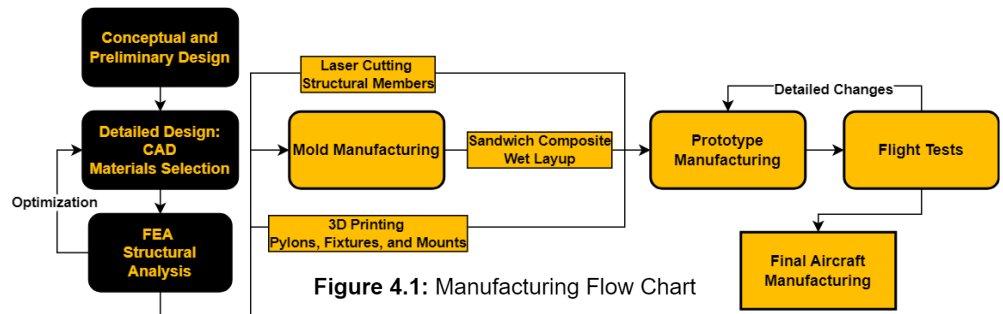


Figure 4.1: Manufacturing Flow Chart

for the aircraft skins will use composite tooling manufactured from CNC-cut foam. All composite components will be manufactured through wet-layup and vacuum bagging processes.

10. Testing Plan

Multiple tests will be run before the construction of the final aircraft, to ensure that our calculations and assumptions hold true. The motor will be run on a test stand both statically and dynamically, to ensure that it produces the calculated amount of thrust. The wing spar will be tested by mounting weights on the tips and measuring the deflection. Once the aircraft is built, landing gear tests will be performed by putting the aircraft in its fully loaded configuration and dropping it from a de red sired height. After testing the radio connection and failsafe reliability, the vehicle will be cleared for flight testing. Taxi and take- off tests will be performed to confirm the calculated values, or identify areas of improvement. Multiple full mission flight tests will then be performed to fine tune the flight performance of the aircraft, and optimize for the score. The glider development will be run separately, with autopilot tests done on a separate test aircraft to remove dependency on other subsystems. Before the full mission test flights, glider release tests will be performed to ensure that the release is smooth and consistently activates the autopilot protocol.

Test	Objective	Method	Sub-System
Preliminary Tests			
Static Thrust Test	Test different motor and prop combinations to verify it produces the expected thrust.	Use a thrust test stand to collect thrust, torque, battery current, and RPM of different motor and propeller combinations.	Propulsion
Dynamic Thrust Test	Collect thrust test stand data in freestream air to get empirical thrust vs. velocity curves.	Mount the test stand in a wind tunnel and collect data on different motor and prop configurations at different airspeeds.	Propulsion
Wing Load Test	Verify maximum load factor of aircraft wing structure to simulate a distributed wing load.	Mount center of aircraft on fixture, load underside of wing with spanwise weights matching lift distribution at max g-loading.	Structures
Wingtip Test	Verify that the aircraft passes the 2.5g competition wing loading requirement.	Load the aircraft with the maximum payload weight and mount it on the wingtips, and measure the deflection.	Structures
Ground Testing			
Landing Gear Stress Test	Verify that the landing gear withstands takeoff and landing shock loads.	Drop the aircraft from an appropriate height. The propeller must not strike the ground at full suspension bottom-out.	Structures/ Mechatronics
Failsafe testing	Test to ensure that the correct commands are given by the receiver during a failsafe.	Turn off the radio transmitter while the aircraft is powered, and check to see that the correct failsafe commands are given.	Avionics
Radio Testing	Ensure that the radio is fully functional, providing adequate data to the receiver.	Verify radio transmitter and receiver communicate at a sufficient range, and at all orientations.	Avionics
In Flight Testing			
Take-Off Speed Test	Record and analyze take-off performance data to optimize performance.	Increase throttle to full, rotate at the calculated rotation speed, and compare the liftoff location to the theoretical values.	Aerodynamics
High Speed Taxi Test	Test the nosewheel and yaw control on the ground, as well as landing gear strength.	Use slow-motion cameras to empirically test that the landing gear holds during future takeoffs.	Mechatronics
Mission Flight Tests	Practice all three flight tests and calculate the scores to identify areas for adjustment.	Perform all three missions and determine if changes to throttle settings, payload weights, etc., can improve scores.	Aerodynamics
Glider Testing			
Glider Detach Test	Test the reliability of the detachment method of the glider from the main aircraft.	Test the detach mechanism to ensure the glider detaches smoothly and the LEDs turn on. Repeat the tests in flight.	Mechatronics
Waypoint Mission Test	Perform gliding waypoint missions to gain familiarity and limit test INAV.	Using a separate test aircraft, perform a waypoint mission and landing similar to the path of the glider in mission 3.	Avionics

Table 5.1: Test Plan Summary